16-Qubit Random Number Generator

Quantum AI (417523A) - SPPU BE AI&DS

CODE

```
import warnings

from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

warnings.filterwarnings("ignore", category=DeprecationWarning)

N_QUBITS = 16
SAMPLES = 8

print("=" * 66)
print("  QUANTUM AI (417523A)  -  PRACTICAL 5")
print("  16-QUBIT RANDOM NUMBER GENERATOR")
print("=" * 66)

qc = QuantumCircuit(N_QUBITS, N_QUBITS)
qc.h(range(N_QUBITS))
qc.measure(range(N_QUBITS), range(N_QUBITS))

sim = AerSimulator()
counts = sim.run(transpile(qc, sim), shots=SAMPLES).result().get_counts()

print(f"\nQubits placed in superposition : {N_QUBITS}")
print(f"Size of the outcome space      : 2^16 = {2 ** N_QUBITS}")
print(f"Each measurement collapses the register onto one of {2 ** N_QUBITS}")
print("equally likely 16-bit strings - genuine hardware randomness.")

print(f"\n{SAMPLES} random 16-bit samples read from the simulator:")
print(f"   {'#':>2}   {'bitstring':<20}{'integer':>8}")
for i, bits in enumerate(counts, 1):
    print(f"   {i:>2}   {bits:<20}{int(bits, 2):>8}")

headline = list(counts)[0]
print("\nHighlighted random 16-bit number :", headline)
print("Grouped as two bytes             :", headline[:8], headline[8:])
print("Decimal value                    :", int(headline, 2))
print("=" * 66)
```

OUTPUT

```
==================================================================
  QUANTUM AI (417523A)  -  PRACTICAL 5
  16-QUBIT RANDOM NUMBER GENERATOR
==================================================================

Qubits placed in superposition : 16
Size of the outcome space      : 2^16 = 65536
Each measurement collapses the register onto one of 65536
equally likely 16-bit strings - genuine hardware randomness.

8 random 16-bit samples read from the simulator:
    #   bitstring            integer
    1   0101101011000110       23238
    2   1111100000110110       63542
    3   0000101001111100        2684
    4   1111101001110111       64119
    5   1100011111010000       51152
    6   1100100001011001       51289
    7   1110011100001100       59148
    8   1011100100110000       47408

Highlighted random 16-bit number : 0101101011000110
Grouped as two bytes             : 01011010 11000110
Decimal value                    : 23238
==================================================================
```

FIGURE: Q5_16qubit_rng

